CP020003

ARTIFICIAL INTELLIGENCE

Khon Kaen University · 2026

Week 07

Association Rules & Market Basket Analysis

Association Rules & Market Basket Analysis **“When customers buy item A, what else do they tend to buy in the same transaction?”**

Instructor: Teerapong Panboonyuen (P'Kao)

https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1

TODAY'S ROADMAP

From Formulas to Modern AI

1. **Why Association Rules?**
   How MBA differs from recommenders
2. **Core Formulas**
   Support, Confidence, Lift — by hand
3. **Worked Example**
   Full numeric walk-through for the exam
4. **Data → Baskets**
   Cleaning & one-hot encoding
5. **Apriori Algorithm**
   Downward closure, step by step
6. **FP-Growth**
   Faster mining, same answer
7. **Item2Vec (Modern AI)**
   Embeddings for basket analysis
8. **Business Impact**
   Turning a rule into revenue

1 · MOTIVATION

Why Association Rule Mining?

Recommenders (Week 6) personalize per user. Association rules (this week) mine co-occurrence across everyone — no user ID required.

	Recommender Systems (Wk 6)	Association Rules (Wk 7)
Unit of analysis	A single user's history over time	A single transaction / basket
Question answered	“What would this person like?”	“What items co-occur together?”
Needs a user ID?	Yes, almost always	No — works on anonymous receipts
Output	A ranked list per user	Rules {A,B} → {C} that apply to anyone
Classic example	Netflix “Because you watched…”	Walmart “Customers also bought…”

Where Market Basket Analysis Shows Up

“Diapers & Beer”

The pedagogically-immortal discovery that launched MBA in retail folklore

Frequently Bought Together

Amazon / Shopee cross-sell widgets at checkout

Shelf Placement

Why milk sits at the back of the store — forcing a longer walk past tempting items

Drug Co-Prescription

Pharmacies flag dangerous medication combinations

Bundle Promotions

Cross-sell email campaigns built from real co-purchase data

Fraud / Anomaly Detection

An unusual item combination can flag a suspicious transaction

The Anatomy of an Association Rule

$$X \Rightarrow Y$$

"If a customer buys itemset **X**, they also tend to buy itemset **Y**" — where X and Y are disjoint (non-overlapping) sets of items.

Notation used throughout this exam

N	Total number of transactions (baskets) in the dataset
$freq(Z)$	Number of transactions that contain itemset Z
$X \Rightarrow Y$	Antecedent X implies consequent Y
$X \cup Y$	The combined itemset — both X and Y present together

Support — “How common is this pattern?”

Out of every transaction in the store, what fraction contains itemset Z?

1

$$\frac{freq(Z)}{N}$$

What it means

- Measures how frequently itemset Z appears across ALL transactions
- Support(Z) always lies between 0 and 1 (or 0% – 100%)
- A high-support itemset is a common shopping pattern
- Used to filter out rare, statistically unreliable combinations via min_support

⚠️ Support alone never tells you direction — Support({Bread, Milk}) does not say whether buying Bread leads to buying Milk, or vice-versa. It only measures joint popularity.

Confidence — “How reliable is this rule?”

Given that a customer bought X, how often did they ALSO buy Y?

2

$$\frac{Support(X \cup Y)}{Support(X)}$$

What it means

- A conditional probability: P(Y bought | X bought)
- Ranges from 0 to 1 (or 0% – 100%)
- Confidence = 1.0 means EVERY basket with X also had Y
- This is the metric mlxtend calls “metric” by default when generating rules

⚠️ Trick #1 (exam favorite): Confidence can be misleadingly high just because Y (e.g. bread) is bought by almost everyone — regardless of X. Confidence alone is NOT enough to call a rule “good.”

Lift — “Is this a REAL association?”

Is X→Y stronger than what random chance would predict, given how popular Y already is?

3

$$\frac{Confidence(X \Rightarrow Y)}{Support(Y)}$$

$$= \frac{Support(X \cup Y)}{Support(X) \cdot Support(Y)}$$

Reading the number

Lift = 1	Independent — pure coincidence, no real association
Lift > 1	Positive association — buying X makes Y MORE likely ✅
Lift < 1	Negative association — buying X makes Y LESS likely ⛔

🔑 Trick #1 (exam favorite): Support alone is not enough. Always check Lift — it is the metric that proves a rule is a genuine pattern, not just a popular item riding along on a high-confidence coattail.

Leverage & Conviction

Two extra metrics you may see in mlxtend output or on the written exam.

L Leverage

$$\text{Support}(X \cup Y) - \text{Support}(X) \cdot \text{Support}(Y)$$

- Lift's “difference” cousin: same idea as Lift, but subtracts instead of divides.
- 0 = independence (instead of 1 for Lift)
- Positive = X and Y co-occur more than chance

C Conviction

$$\frac{1 - Support(Y)}{1 - Confidence(X \Rightarrow Y)}$$

- = ∞ when the rule has PERFECT confidence (always true)
- The larger the value, the more the rule “beats” independence
- Like Lift, but sensitive to the direction of the rule

The Toy Dataset: Six Transactions

Six transactions from a small shop. We will evaluate the rule {Diaper} ⇒ {Beer} by hand, step by step — exactly as you would on the written exam.

Transaction	Items Purchased
T1	Bread, Milk
T2	Bread, Diaper, Beer, Eggs
T3	Milk, Diaper, Beer, Cola
T4	Bread, Milk, Diaper, Beer
T5	Bread, Milk, Diaper, Cola
T6	Bread, Milk, Beer

$N = 6$

total transactions

Question to answer:

Evaluate the rule

{Diaper} ⇒ {Beer}

Find: Support, Confidence, Lift

Count Occurrences of Each Item

Scan all 6 transactions and mark which ones contain Diaper, Beer, or both.

Txn	Items	Diaper?	Beer?
T1	Bread, Milk	—	—
T2	Bread, Diaper, Beer, Eggs	✓	✓
T3	Milk, Diaper, Beer, Cola	✓	✓
T4	Bread, Milk, Diaper, Beer	✓	✓
T5	Bread, Milk, Diaper, Cola	✓	—
T6	Bread, Milk, Beer	—	✓

freq(Diaper)

in: T2, T3, T4, T5

$= 4$

freq(Beer)

in: T2, T3, T4, T6

$= 4$

freq(Diaper, Beer)

in: T2, T3, T4

$= 3$

Calculate Support

Support(Z) = freq(Z) / N — substitute the counts from Step 1 (N = 6)

Support(Diaper)

$$\frac{4}{6}$$

= 0.667

Support(Beer)

$$\frac{4}{6}$$

= 0.667

Support(Diaper, Beer)

$$\frac{3}{6}$$

= 0.5

💡 Reading the numbers: Diaper alone appears in 66.7% of all baskets, Beer alone in 66.7%, and the two together in exactly half of all baskets. These three numbers feed directly into Confidence and Lift on the next slides.

Calculate Confidence

Confidence(X⇒Y) = Support(X∪Y) / Support(X) — substitute the Support values from Step 2

$$Confidence(Diaper \Rightarrow Beer)$$

$$\frac{0.5}{0.667}$$

$$= 0.75$$

💡 Interpretation: 75% of transactions that contained Diaper ALSO contained Beer. That is a strong conditional signal — but remember Trick #1: high confidence could still be inflated if Beer itself is just a very popular item. We must check Lift to be sure.

Calculate Lift

Lift(X⇒Y) = Confidence(X⇒Y) / Support(Y) — substitute the Confidence value from Step 3

$$Lift(Diaper \Rightarrow Beer)$$

$$\frac{0.75}{0.667}$$

$$= 1.125$$

✅ Final verdict: Lift = 1.125 > 1, so buying Diaper makes buying Beer about 12.5% MORE likely than random chance. This confirms {Diaper}⇒{Beer} is a real, if mild, positive association — not just Beer's general popularity riding along.

Calculate Leverage & Conviction

Using the same Support and Confidence values computed above

Leverage

$$0.5 - (0.667 \times 0.667)$$

= 0.056

Conviction

$$\frac{1 - 0.667}{1 - 0.75}$$

= 1.333

📝 Full answer for the exam: Support = 0.5, Confidence = 75%, Lift = 1.125, Leverage = 0.056, Conviction = 1.333. All five numbers should be reported together — each tells a slightly different part of the story.

✍ TRY IT YOURSELF

Practice Before the Real Exam

Using the same six toy transactions, calculate Support, Confidence, and Lift by hand for the rule:

$$\{\text{Bread}\} \Rightarrow \{\text{Milk}\}$$

1. Count freq(Bread), freq(Milk), freq(Bread ∩ Milk)
2. Compute Support for each
3. Compute Confidence(Bread⇒Milk)
4. Compute Lift and interpret it
5. Verify your work in mlxtend once you code it

📚 Association Rule — Formula Review

Let:

- A = antecedent (the item/items on the left)
- B = consequent (the item on the right)
- N = total number of transactions

1. Support

For an item/itemset:

$$\boxed{\text{Support}(A) = \frac{\text{freq}(A)}{N}}$$

For two itemsets occurring together:

$$\boxed{\text{Support}(A, B) = \frac{\text{freq}(A, B)}{N}}$$

2. Confidence

$$\boxed{\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A, B)}{\text{Support}(A)}}$$

An equivalent formula using frequencies is:

$$\boxed{\text{Confidence}(A \Rightarrow B) = \frac{\text{freq}(A, B)}{\text{freq}(A)}}$$

3. Lift

$$\boxed{\text{Lift}(A \Rightarrow B) = \frac{\text{Confidence}(A \Rightarrow B)}{\text{Support}(B)}}$$

or equivalently:

$$\boxed{\text{Lift}(A \Rightarrow B) = \frac{\text{Support}(A, B)}{\text{Support}(A) \times \text{Support}(B)}}$$

Interpretation:

- Lift > 1 → positive association
- Lift $= 1$ → no association / independent
- Lift < 1 → negative association

3. Lift

$$\boxed{\text{Lift}(A \Rightarrow B) = \frac{\text{Confidence}(A \Rightarrow B)}{\text{Support}(B)}}$$

or equivalently:

$$\boxed{\text{Lift}(A \Rightarrow B) = \frac{\text{Support}(A, B)}{\text{Support}(A) \times \text{Support}(B)}}$$

Interpretation:

- Lift > 1 → positive association
- Lift $= 1$ → no association / independent
- Lift < 1 → negative association

---

4. Leverage

$$\boxed{\text{Leverage}(A \Rightarrow B) = \text{Support}(A, B) - \text{Support}(A) \times \text{Support}(B)}$$

---

5. Conviction

$$\boxed{\text{Conviction}(A \Rightarrow B) = \frac{1 - \text{Support}(B)}{1 - \text{Confidence}(A \Rightarrow B)}}$$

🍜 Association Rule Practice — Thai Food

Question 1 — Easy

A Thai restaurant records the following 6 customer orders:

Transaction	Items
T1	Som Tam, Sticky Rice
T2	Som Tam, Grilled Chicken
T3	Pad Thai, Grilled Chicken
T4	Som Tam, Grilled Chicken
T5	Tom Yum, Sticky Rice
T6	Som Tam, Sticky Rice

🍜 Association Rule Practice — Thai Food

Question 1 — Easy

A Thai restaurant records the following 6 customer orders:

Transaction	Items
T1	Som Tam, Sticky Rice
T2	Som Tam, Grilled Chicken
T3	Pad Thai, Grilled Chicken
T4	Som Tam, Grilled Chicken
T5	Tom Yum, Sticky Rice
T6	Som Tam, Sticky Rice

Evaluate the rule:

$$\{\text{Som Tam}\} \Rightarrow \{\text{Grilled Chicken}\}$$

Calculate:

1. Frequency of Som Tam
2. Frequency of Grilled Chicken
3. Frequency of Som Tam and Grilled Chicken
4. Support of the rule
5. Confidence
6. Lift

🍜 Association Rule Practice — Thai Food

Question 1 — Easy

A Thai restaurant records the following 6 customer orders:

Transaction	Items
T1	Som Tam, Sticky Rice
T2	Som Tam, Grilled Chicken
T3	Pad Thai, Grilled Chicken
T4	Som Tam, Grilled Chicken
T5	Tom Yum, Sticky Rice
T6	Som Tam, Sticky Rice

Evaluate the rule:

$$\{\text{Som Tam}\} \Rightarrow \{\text{Grilled Chicken}\}$$

Calculate:

1. Frequency of Som Tam
2. Frequency of Grilled Chicken
3. Frequency of Som Tam and Grilled Chicken
4. Support of the rule
5. Confidence
6. Lift

NOTES

🍜 Association Rule Practice — Thai Food

Question 1 — Easy

A Thai restaurant records the following 6 customer orders:

Transaction	Items
T1	Som Tam, Sticky Rice
T2	Som Tam, Grilled Chicken
T3	Pad Thai, Grilled Chicken
T4	Som Tam, Grilled Chicken
T5	Tom Yum, Sticky Rice
T6	Som Tam, Sticky Rice

Evaluate the rule:

$$\{\text{Som Tam}\} \Rightarrow \{\text{Grilled Chicken}\}$$

Calculate:

1. Frequency of Som Tam
2. Frequency of Grilled Chicken
3. Frequency of Som Tam and Grilled Chicken
4. Support of the rule
5. Confidence
6. Lift

NOTES

🍜 Association Rule Practice — Thai Food

Question 1 — Easy

A Thai restaurant records the following 6 customer orders:

Transaction	Items
T1	Som Tam, Sticky Rice
T2	Som Tam, Grilled Chicken
T3	Pad Thai, Grilled Chicken
T4	Som Tam, Grilled Chicken
T5	Tom Yum, Sticky Rice
T6	Som Tam, Sticky Rice

Evaluate the rule:

$$\{\text{Som Tam}\} \Rightarrow \{\text{Grilled Chicken}\}$$

Calculate:

1. Frequency of Som Tam
2. Frequency of Grilled Chicken
3. Frequency of Som Tam and Grilled Chicken
4. Support of the rule
5. Confidence
6. Lift

NOTES

Cleaning Transaction Logs → Basket Format

Real data (KKU Online Retail) starts as one row PER ITEM. Apriori / FP-Growth need one row PER TRANSACTION with the set of items inside.

1. **Drop cancellations**
   InvoiceNo starting with “C” = a return, not a purchase
2. **Drop bad rows**
   Quantity ≤ 0 or UnitPrice ≤ 0 = refunds / data errors
3. **Drop missing names**
   Rows with no Description are dropped
4. **Strip whitespace**
   “HEART ” vs “HEART” would count as 2 products

groupby(InvoiceNo) → one-hot encode

InvoiceNo	Bread	Milk	Diaper	Beer
T1	1	1	0	0
T2	1	0	1	1
T4	1	1	1	1

Every positive quantity → 1 (present). Everything else → 0 (absent). Apriori cares whether an item WAS bought — not how many units.

Apriori — The Downward Closure Property

The problem: with M unique products there are 2^M-1 possible itemsets. With just 3,000 products, that is more itemsets than atoms in the universe.

🔑 **The key insight (Apriori Property)**

If an itemset is INFREQUENT, every superset of it is guaranteed to be infrequent too.
Equivalently: every subset of a FREQUENT itemset must itself be frequent.

This lets Apriori prune the search space aggressively:

1. **Find frequent 1-itemsets**
   Single items with support ≥ min_support

2. **Generate 2-itemset candidates**
   Only from frequent 1-itemsets; keep the frequent ones

3. **Generate 3-itemset candidates**
   Only from frequent 2-itemsets; repeat…

4. **Stop**
   When no new frequent itemsets can be generated

Apriori in Practice — Two Function Calls

A

Step A — find frequent itemsets

```
apriori(basket_encoded,
  min_support=0.03,
  use_colnames=True)
```

An itemset must appear in ≥ 3% of transactions to “pass” and move to the next round of candidate generation.

B

Step B — turn itemsets into rules

```
association_rules(
  frequent_itemsets,
  metric="lift", min_threshold=1.0)
```

Splits every frequent itemset into every antecedent → consequent pair, computing Support / Confidence / Lift for each.

🔑 **Trick #3 — Filter, don't just sort.** A rule {X} → {most popular item} always has sky-high Confidence for a boring reason. Always require lift > 1, and typically also a minimum support, before calling a rule “interesting.”

FP-Growth — Same Answer, No Candidate Generation

Apriori's weakness: it re-scans the ENTIRE dataset once per itemset size, generating huge numbers of candidates that mostly turn out infrequent — wasted work.

📊 **Pass 1: Count support**

Count every single item's support; discard infrequent items

🌳 **Pass 2: Build the FP-tree**

Insert transactions into a prefix tree, items ordered by frequency — shared items share tree branches (this is what makes it compact)

⛏️ **Pass 3: Mine recursively**

Divide-and-conquer mining directly on the tree — no re-scanning raw data, no candidate generation

✅ **Output is mathematically IDENTICAL to Apriori — only the algorithm used to get there differs. FP-Growth is simply much faster on large, dense datasets.**

Apriori vs. FP-Growth

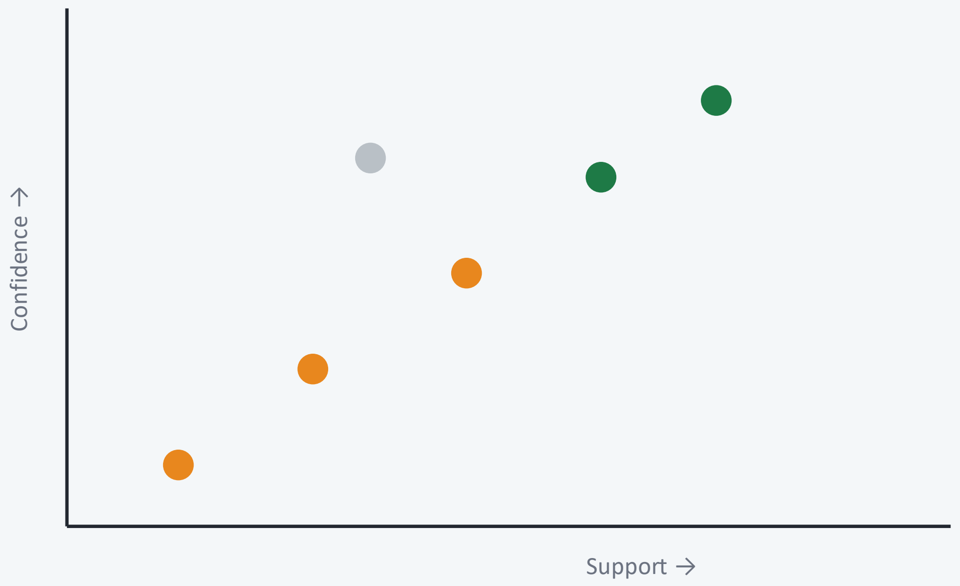
	Apriori	FP-Growth
Passes over data	One pass per itemset size (many)	Just 2 passes total
Candidate generation	Yes — generates then tests candidates	None — mines the tree directly
Memory structure	Flat candidate lists	Compact FP-tree (shared branches)
Speed on small data	Fast enough	Comparable, sometimes marginally faster
Speed on large/dense data	Slows sharply — combinatorial cost	Scales far better — minutes vs. hours
Frequent itemsets found	Identical set of itemsets	Identical set of itemsets

🔑 Trick #4 — On a small dataset the timing gap is barely noticeable. Re-run both on the full UK dataset (thousands of invoices) and the gap grows dramatically.

Visualizing Association Rules

Scatter Plot

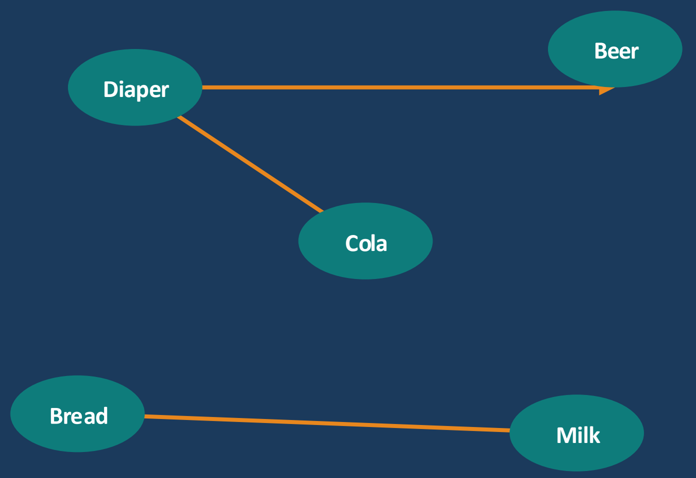
Support (x-axis) vs Confidence (y-axis), colored by Lift



Goal: spot the sweet spot — frequent, confident, AND genuinely associated (bright dots).

Network Graph

Nodes = products. Arrow A→B means "buying A implies buying B"



Mirrors how a manager thinks about physical shelf placement.

Item2Vec — Embedding-Based Basket Analysis

Apriori/FP-Growth are exact, symbolic methods — a rule either clears min_support or it doesn't. Item2Vec borrows Word2Vec's idea from NLP instead.

NLP concept (Word2Vec)		Basket Analysis (Item2Vec)
“Sentence”	→	One customer's basket (list of StockCodes)
“Word”	→	One product (StockCode)
Words in similar contexts have similar meaning	→	Products bought alongside similar products get similar embeddings

```
gensim.Word2Vec(sentences=basket_sequences, vector_size=32, window=10, sg=1, epochs=20)
```

No explicit support/confidence table is ever computed — similarity comes purely from the learned vectors.

🔑 Trick #5 — Rules give an interpretable, auditable statement (“87% confidence, 3.2x lift”). Embeddings give a continuous similarity score that scales to millions of products but doesn't explain WHY. Production systems often run both.

Apriori vs. FP-Growth vs. Item2Vec

	Apriori	FP-Growth	Item2Vec
Output	Exact rules w/ Support/Conf/Lift	Same as Apriori, faster	Dense vector per product
Interpretability	★★★★★ Fully auditable	★★★★★ Fully auditable	★★ “Similar” but not “why”
Scales to huge catalogs?	⛔ Combinatorial explosion	⚠️ Better, still limited	✅ Scales to millions of items
Needs min_support tuning?	Yes	Yes	No (needs vector_size, window)
Captures sequence/order?	No — basket is a set	No — basket is a set	Loosely, via context window

🔑 *Trick #6 — Fancier is not always better. A well-tuned Apriori/FP-Growth pipeline often OUTPERFORMS a complex neural approach on small-to-medium retail data, and is far easier to explain to a stakeholder. Try the simple, auditable baseline first.*

PRACTICAL TRICKS & PITFALLS

Every Practitioner Should Know These

min_support too low

Combinatorial explosion — millions of near-meaningless itemsets, extremely slow

min_support too high

You only rediscover "bread and milk" — obvious, useless rules

Ignoring Lift

High-confidence rules that are just "→ most popular item," not a real pattern

Forgetting cancellations/returns

Phantom "purchases" that were actually refunded pollute the basket

Mixing countries/stores

Regional buying habits cancel out into bland, generic rules

Treating quantity as meaningful

1 vs. 50 units of the same item look identical to presence/absence encoding

Turning a Rule Into Estimated Revenue

A rule sitting in a notebook makes no money. Let's cost out one concrete rule — the same way Week 6 costed out a recommender.

1	**Antecedent transactions**	`antecedent_support × N`	e.g. 0.20 × 200 = 40 baskets contain the antecedent
2	**Already buy both**	`support × N`	e.g. 0.08 × 200 = 16 baskets already buy both items
3	**Eligible non-buyers**	`antecedent_txns − already_buy_both`	40 − 16 = 24 baskets are eligible for a prompt
4	**Expected new sales**	`eligible × uplift rate`	24 × 10% take-rate = 2.4 new units sold
5	**Expected incremental revenue**	`new_sales × avg_price`	2.4 × £5.00 = £12.00 per rule, per period

✅ SUMMARY

Key Takeaways

1. Association rules mine co-occurrence across transactions — not individual user preference (complementary to Week 6's recommenders)

2. Three numbers matter for the exam: Support (how common), Confidence (how reliable), Lift (how genuinely associated) — always compute all three

3. Real data must become basket format: drop cancellations, group by invoice, one-hot encode (never raw quantities)

4. Apriori prunes the search space via downward closure: no superset of an infrequent itemset can be frequent

5. FP-Growth finds the SAME frequent itemsets faster, using a compact tree and no repeated candidate generation

6. Always filter rules by Lift, not just Confidence, to avoid mistaking popularity for a genuine pattern

7. Item2Vec brings a modern, embedding-based alternative that scales further at the cost of interpretability

8. Every rule should end in a business action — shelf placement, bundling, or a targeted promo

📝 HOMEWORK THIS WEEK

Practice Before You Submit

1. **Change COUNTRY** — Try “Germany” or the full “United Kingdom.” Does min_support = 0.03 still make sense?
2. **Re-time Apriori vs. FP-Growth** — On the full UK dataset. Plot runtime vs. min_support for both algorithms.
3. **Hand-calculate a new rule** — Support, Confidence, Lift for {Bread}⇒{Milk} — show your work as on a written exam.
4. **Tune Item2Vec** — Try vector_size=64, window=5. Why might a smaller window give more relevant neighbors?
5. **(Challenge) Embedding-only rule** — Find 2 rare, long-tail products that Item2Vec links but Apriori would miss at any reasonable min_support.

Turn in: your completed notebook (all cells run top-to-bottom, no errors), with written answers to Q1, Q3, and Q5.